

Your Benchmark Is Not Saturated: Reviving Multiple-Choice Evaluation with Answer Pooling

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Abstract

Multiple-choice benchmarks are the backbone of language model evaluation because they are cheap and exactly gradable, yet frontier models now crowd 90% on them, and the standard remedy, authoring harder items, is expensive, slow, and restarts the contamination clock. We propose answer pooling: pool the options of N questions that share a document or topic into one exclusive candidate list and ask the model for the complete assignment. Deleting gold answers from the pool further manufactures unanswerable questions at any ratio, so abstention is scored jointly with accuracy in the same forward pass, with no judge and no new annotation. On five benchmarks spanning long-form reading, school and expert exams, general knowledge, and Chinese, the transform is uniformly harder than item-paired MCQ with model rankings preserved: on QuALITY the best model falls from 0.91 to 0.52 while the guessing floor drops from 0.25 to 5.4×10^{-7} . Withholding answers exposes what accuracy cannot see: two models with statistically tied accuracy differ by 0.42 in false-answer rate, and two open model families answer 92 to 100% of unanswerable questions at every size from 7B to 32B, where scaling improves compliance with a written instruction to decline but never the disposition to decline under competition. Most surprising, the measured rate belongs to the instrument as much as to the model: printing a none option halves false answers, multiplies false abstentions by 2 to 4, and flips a significant accuracy ranking. Any benchmark with several questions per context can be revived and audited this way for free, and abstention results are comparable only under a stated response format.

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🔗 **Code:** <https://github.com/AbobakerAhmed/AnswerPool>

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1 Introduction

Multiple-choice question answering (MCQA) is the workhorse of language model evaluation because it is cheap, judge free, and exactly gradable. It is also running out of room. Frontier models crowd 90% on established benchmarks, and once scores compress against the ceiling a benchmark stops ordering models. The standard remedy is to author harder items: MMLU-Pro [1] writes new distractors with LLM assistance and validation passes, and still lands at a guessing floor of 0.1. Authoring is expensive, must be repeated per benchmark, and every fresh release restarts the contamination clock.

The treadmill rests on a premise, that a saturated benchmark is an exhausted one. We argue the scores are saturated but the labels are not, and the difference is the question asked of them. Take N questions that share a document. Pool their options into one shared candidate list of M entries. Ask the model to assign every question its answer, letters being mutually exclusive. Nothing is authored, the labels are untouched, and any benchmark with several questions per context converts for free.

The effect on headroom is large (Table 3). On QuALITY [2], filtered of items solvable closed book, the best evaluated model drops from 0.9104 (MCQ accuracy) to 0.5200 (exact assignment), the guessing floor falls from 0.25 to 5.4×10^{-7} , and best-to-worst model separation doubles from 0.20 to 0.44. Three properties make this difficulty defensible rather than decorative. Rankings are preserved while gaps widen, scaling inversely with model strength. A choices-only baseline is provably capped (Proposition 1) and measures at chance on QuALITY. And the group-level closed-book rate is about 1%, matching a group-scored MCQ control, so the restored headroom is not shortcut headroom.

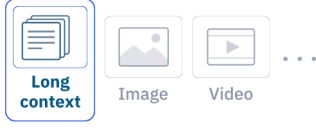
The shared pool then pays a second time. Every MCQ item silently promises the model that the answer is present, a guarantee deployment routinely breaks. Because our candidates are pooled and exclusive, we can withhold the gold for a fraction p of questions: those questions have no correct answer, we know exactly which, and it cost nothing. Deletion itself is not unique to the pool. Replacing the gold of a standard MCQ item with a printed none option manufactures unanswerability just as cheaply, and we build exactly that as our second instrument. What the pool changes is the affordance: abstention must be produced against roughly 17 live on-topic candidates that other questions compete for, rather than selected from a printed escape option, and the same construction is what de-saturates the benchmark, which delete-gold MCQ does not. The two instruments form a ladder of increasing selection pressure, and running both rungs on identical items shows the ladder does not converge to one number, which is itself the finding. Confabulation is rampant and invisible to accuracy. Two models whose accuracy is statistically tied differ by 0.42 in false-answer rate (95% CI [+0.375, +0.465]). Our weakest model declined 2 of 900 unanswerable items and 0 of 2,100 answerable ones. And the measured rate is strongly instrument dependent: printing a none option halves confabulation, multiplies over-abstention by 2 to 4 for every model, and flips a significant accuracy ranking between two frontier-family models.

Our contributions are methodological:

- **Answer pooling, a zero-authoring de-saturation transform.** N co-document questions compete for one exclusive candidate list, scored per question and by exact assignment. Two guarantees come with the construction: partial-input accuracy is provably capped at $1/N$, reaching the $1/M$ floor on style-homogeneous pools (Proposition 1), and the guessing floor is $(M-N)/M!$ (Proposition 2), with N and M as difficulty dials on unchanged labels.
- **Withheld golds, unanswerability without annotation.** Deleting a gold from the pool

Input (MCQs Benchmark)

Context (depends on the benchmark type)



Questions & their choices

Q1	A1	B1	C1	D1
Q2	A2	B2	C2	D2
Q3	A3	B3	C3	D3
Q4	A4	B4	C4	D4

1. Filtering

Drop questions solvable without the context

Q4 dropped

2. Withholding

With a fixed ratio p , replace the correct choices with None

B2 → None

3. Pooling + Shuffling

All choices of the kept questions

Our Formulation (Matching Task)

List A

Q1
Q2
Q3

List B

C2
B1
D3
A3
A2
C1
...
None

The model must assign **every** question in List A to exactly **one** entry in List B.

Figure 1 The transform. Stage 1: benchmarks ship a passage and N questions, each with one gold answer (green, check) and three distractors (gray). Stage 2: all options are pooled into one exclusive candidate list of M entries, shuffled, and a fraction p of golds is withheld (scissors, dashed slot). Stage 3: a frozen language model reads the passage, the questions, and the pool, and produces an injective assignment, letters used at most once. A question whose gold was withheld is correctly answered only by none (red, dashed). Shown with $N=3$ and $M=8$ for legibility, experiments use $N=5$ and $M \approx 20$.

makes its question unanswerable with exact ground truth, at any ratio p , from labels that already exist. No annotator writes an unanswerable question.

- **Judge-free abstention metrics from the accuracy pass.** False-answer rate, false-abstention rate, and strict exact assignment, a single joint number for competence and calibration, all exact-match scored in the same forward pass that produces accuracy.
- **An affordance-controlled protocol and toolkit.** The same items with identical withheld assignments evaluated under varied response affordances (competitive assignment, None-option MCQ, free generation), with the infrastructure that makes the comparison trustworthy: grammar-constrained decoding for full format compliance, a closed-book filter derived only from closed-book runs, and group-resampled paired bootstraps. Released as a conversion pipeline.

The empirical findings these instruments produce, restored headroom with preserved rankings, confabulation invisible to accuracy, and abstention rates that belong to the instrument as much as to the model, are previewed above and quantified in Section 4.

2 Related Work

Benchmark saturation and revival.. MMLU-Pro [1] de-saturates MMLU by authoring ten-option items with LLM-generated distractors and validation passes, reaching a 0.1 floor. BeyondBench [3] transforms existing benchmarks, but targets contamination rather than headroom. Both lines either author new content or leave the guessing floor high. Reviving a benchmark without authoring anything, at floors below 10^{-3} , remains open.

MCQ artifacts.. Balepur et al. [4] show that models beat a majority baseline on choices alone in 11 of 12 dataset and model pairs, with gains up to +0.33, and trace the worst cases to stylistic cues separating human-written golds from model-written distractors. They call for format-level fixes

Table 1 Capability comparison. Tilde marks partial satisfaction. Ours is the only approach that is simultaneously zero-authoring, low-floor, and affordance-controlled.

	zero new authoring	floor < 10^{-3}	abstention measurable	acc+abst same run	adversarial candidates free	affordance controlled	black-box	judge-free
MCQ	✓	×	×	–	×	×	✓	✓
Group-scored MCQ	✓	~	×	–	×	×	✓	✓
MMLU-Pro [1]	×	×	×	–	×	×	✓	✓
BeyondBench [3]	~	×	×	–	×	×	✓	✓
SQuAD 2.0 [7]	×	×	✓	×	×	×	✓	✓
AbstentionBench [8]	×	×	✓	×	×	×	✓	~
None-option MCQ (NOTA)	✓	×	✓	✓	×	×	✓	✓
Probes	✓	×	✓	✓	–	×	×	✓
Free-form + judge [10]	✓	×	✓	×	–	×	✓	×
Ours	✓	✓	✓	✓	✓	✓	✓	✓

such as hand-authored Winograd pairs [5]. Zheng et al. [6] identify option-identifier token bias and correct it post hoc. These works diagnose or debias. None restores headroom, and a format that provably resists the choices-only attack has not been available.

Unanswerability and abstention.. SQuAD 2.0 [7] pays annotators to write unanswerable questions. AbstentionBench [8] curates inherently unanswerable queries into a separate benchmark and finds scaling does not fix abstention. None-of-the-above studies add an escape option to single MCQ items. SATA-Bench [9] moves to select-all-that-apply. All of these measure abstention under exactly one response affordance and require either authored labels or a separate evaluation run. Manufacturing unanswerability for free from existing labels, scoring it jointly with accuracy in the same run, and holding items fixed while varying the affordance are all missing. The last gap matters most: without it, the affordance artifact we report in Section 4.3 is invisible.

Judges and probes.. Free-form evaluation with an LLM judge is the validity anchor for abstention [10] but costs a judge call per item and inherits judge bias. Hidden-state abstention probes require open weights. An exact-match, black-box, judge-free abstention measure has not been available.

3 Method

3.1 The transform

Let a document carry questions $q_1, \dots, q_N$ with gold answers $g_1, \dots, g_N$ and authored distractors. We pool the golds together with each question’s own distractors into one candidate list of $M \approx 4N$ entries, deduplicated and shuffled. The model reads the document, all N questions, and the pool, and outputs one letter per question. Letters are mutually exclusive, so the output is an injective assignment.

Two scores are reported. **Per-question accuracy** (floor $1/M$) gives partial credit and statistical power. **Exact assignment** requires every slot correct and carries the paper’s headline claim. The de-saturation result lives at the exact-assignment level, and we say so explicitly: per-question accuracy barely moves (0.8727 versus 0.9104 for the best model). The honest comparator for a conjunctive score is therefore **group-scored MCQ**, known as testlet scoring in the measurement literature, which scores N standard MCQ items as one all-or-nothing unit, and it appears inside our main table rather than in an appendix.

N , M , and the withhold fraction p (Section 3.2) are three difficulty dials operating on the same annotations.

Proposition 1 (Partial-input cap). *The candidate pool is invariant to the identity and order of the questions, so an adversary shown only the pool learns nothing about which question owns which candidate. Expected per-question accuracy is therefore at most $1/N$ even if golds are perfectly distinguishable from distractors (the adversary can at best identify the gold set and guess the pairing), and equals $1/M$ when candidates are stylistically exchangeable.*

MCQ carries no such cap: choices-only accuracy up to 0.585 has been measured [4] and 1.0 is possible in principle. The gap between the $1/M$ floor and the $1/N$ cap makes the choices-only arm an *artifact detector*: accuracy at $1/M$ certifies a style-homogeneous pool, and any excess, bounded by $1/N$, quantifies gold–distractor separability in the source benchmark.

Proposition 2 (Exact-assignment floor). *Uniform guessing over injective assignments succeeds with probability $\frac{(M-N)!}{M!}$, which at $N=5, M=20$ is 5.4×10^{-7} , six orders of magnitude below the MCQ floor of 0.25 and three below group-scored MCQ at $4^{-5} \approx 9.8 \times 10^{-4}$.*

3.2 Withholding: the axis the pool unlocks

Every MCQ item guarantees the model that the answer is present. Because our candidates are pooled and exclusive, we withdraw that guarantee for a fraction p of questions by deleting their golds from the pool. The model may answer none for any question. Ground truth is exact and free, since we know which golds were deleted. Metrics: the **false-answer** rate, answering a question whose gold is absent (confabulation), the **false-abstention** rate, answering none when the gold is present (over-refusal), and **strict exact assignment**, every slot correct and none on every withheld slot, a single joint number for competence and calibration that accuracy cannot express.

3.3 Well-formedness of the transform

The source benchmark certifies each question individually, QuALITY items were validated by annotators with the passage. Pooling can therefore break solvability only through interactions between items, and the criteria for a well-formed group are exactly the interaction failures:

- **W1, unique golds.** No two golds in a group are semantically equivalent. String-level duplicates are rejected at build time, semantic near-duplicates require the screen below.
- **W2, unique correctness.** Every candidate is a defensible answer to at most one question in the group.
- **W3, question independence.** No question’s text reveals another question’s answer.
- **W4, withhold soundness.** After a gold is deleted, no remaining candidate is defensible for its question, so none is uniquely correct. W4 follows from W1 and W2, so screening the first two certifies the withhold arm. A W4 violation would overcount the false-answer rate, which is why this corollary matters.

Three instruments verify the criteria. **Verifier screen**: for each question a screener model sees the passage, the question, the pool, and the gold revealed, and answers one binary, whether any other candidate is also fully correct. Flagged by at least 2 of 3 screeners on QuALITY: 7.5% of questions (36 naming another question’s gold, 77 naming a distractor), touching 83 of 300 groups, a retention of 72.3%. On the screened subset every headline number holds or improves, exact assignment for the best model rises from 0.520 to 0.562, the accuracy-tied pair stays tied, and its false-answer gap grows to +0.438, so the ceiling is capability, not ambiguity. On MMLU-Pro, retention is 83.5% and only 2 of 1200 questions have another question’s gold as a second valid answer, so numeric answers collide far less than expected, and the separability measured in

Section 4.5 survives screening unchanged, confirming that the partial-input leak is gold–distractor separability rather than ambiguity. **Easy-matching control:** pools built from the group’s golds plus golds of unrelated articles are topically separable. Every model tested, six models across two families and three scales, *exceeds its own MCQ accuracy* there (per-question 0.975 versus 0.910 for the best model, 0.761 versus 0.701 for the weakest and oldest 7B), so format assembly costs nothing at any capability level tested and the entire drop on real pools is discrimination among plausible same-document candidates. **Exclusivity audit:** the decoding grammar permits none on every line (two withheld golds require two none answers, and the withheld count is never revealed to the model), but distinct letters are requested rather than grammar-enforced. Measured on the held-out arms, 2.3 to 3.0% of groups contain a repeated letter with the full pool, and 1.3 to 6.3% under withholding. Repeated letters are scored as ordinary errors, at most one of the tied slots can be correct and exact assignment fails automatically. The per-model rates under withholding (0.013, 0.020, 0.047, 0.063) follow the same ordering as the false-answer rate. Withholding roughly doubles the duplication rate for the two weakest abstainers (pooled 2.8% to 5.5%) while the strongest abstainer’s rate falls (2.3% to 1.3%), so duplication is a second expression of the disposition to answer. Slot-level targeting is not established: duplicates involve a withheld slot in 33 of 43 cases, close to the 70% chance baseline when two of five slots are withheld. When they do, the shared letter is the other question’s own gold at roughly the model’s accuracy rate (21 of 33), so the typical duplicate is a withheld question repeating an answer already claimed elsewhere rather than abstaining.

3.4 Instrument arms

To measure whether the reported abstention rate depends on how abstention is offered, we hold the items fixed and vary only the response affordance.

Competition (matching). As above. A single prose line permits writing none. The model must produce it against roughly 17 live on-topic candidates, with no printed escape option in any list.

Prose escape (Prose-MCQ). The decomposition arm. Delete-gold MCQ at a constant three options with no printed escape, only the same one-line prose permission the matching arm uses. Matching and None-option MCQ differ in two ways at once (competitive pool versus independent items, prose permission versus printed option), and this middle arm attributes the affordance effect to its components.

Offered escape (None-option MCQ). Delete-gold MCQ. The same questions with their original options, where a none option is printed as a lettered choice. Three invariants are load-bearing. The none option appears in every item, otherwise its presence signals the answer. The option count is constant at four, with one distractor dropped when the gold is kept, otherwise the count leaks the label. Withheld questions mirror the matching arm item for item.

Generation (free-form). The same questions asked open-ended with no abstention instruction, since an instruction would rebuild the offered escape in prose. Unanswerability is a mismatched passage, the deployment case of a retrieval miss. A judge classifies each response as ANSWER or ABSTAIN only, never correctness, keeping the judgement narrow and reliable.

All three arms share one item set with identical withheld assignments, aligned by question text and verified identical (2,070 of 2,070 items).

3.5 Experimental setup

Data. Five benchmarks, split by whether a passage exists (Table 2). QuALITY [2] is the primary one and carries the instrument comparisons. RACE [11] is the second open-book corpus, and

Table 2 The five benchmarks. Open-book benchmarks ship a passage and support every instrument arm, including the affordance and generation comparisons. The closed-book three ship no passage, so those arms do not apply and groups are formed by topic rather than by document. Closed-book solvability is the fraction of questions that at least 2 of 3 models answer correctly with the passage withheld, which is undefined where no passage exists.

	context	domain	N, M	groups	closed book
<i>Open book (reading comprehension)</i>					
QuALITY	~5k tokens	literary narrative	5, 20	300	49.7%
RACE	~300 words	English exams	4, 16	95	—
<i>Closed book (knowledge)</i>					
MMLU-Pro	none	academic, 10-option	3, 24	400	n/a
GPQA	none	expert STEM	5, 20	83	n/a
C-Eval	none	Chinese subject exams	5, 20	237	n/a

its passages are an order of magnitude shorter, which lets us separate effects of the response format from effects of long-context load. MMLU-Pro [1], GPQA and C-Eval are closed book, group by topic instead of document, and exercise the transform where the pool’s neighbours are topic-mates rather than same-document siblings. On MMLU-Pro the ten-option items would overflow the 26-letter answer alphabet, so 7 of each question’s 9 distractors are sampled into the pool.

Closed-book filter. On the two open-book benchmarks, items that at least 2 of 3 models answer without the passage are removed before any evaluation. Filtering on closed-book runs is independent of every open-versus-closed-book comparison made afterwards. The residual is a property of the data rather than one family’s pretraining: after filtering QuALITY, Qwen3-8B scores 0.334 closed book against Gemini’s 0.349 and 0.399, a family the filter never saw. RACE is markedly leakier than QuALITY, 0.60 to 0.71 closed book before filtering against 0.47 to 0.54, and stays at 0.47 to 0.60 afterwards, so a 2-of-3 vote cannot remove all the signal that individual models still exploit there.

Models. Gemini 3.6 Flash, Gemini 3.1 Flash-Lite, Gemma 4 26B (Vertex API), and locally via vLLM: Qwen2.5-7B, Qwen3-8B, Qwen3-32B (FP8), Gemma 3 12B, and Gemma 3 27B.

Constrained decoding. Local models decode under an xgrammar-enforced output grammar, making format compliance 100% by construction: no unparsable responses on any reported arm. An unconstrained control changes the false-answer rate by 0.004 and costs 101 unparsable responses, so the constraint does not move the measure. Qwen3 additionally requires thinking mode disabled.

Statistics. All contrasts are paired bootstraps resampling groups (2,000 resamples, 95% CIs). Model-versus-model contrasts are paired at the item level, every model answers the same file. Arm-versus-arm contrasts are paired at the item level for mirror-built arms and at the article-group level otherwise.

Reproducibility. Every reported number regenerates deterministically from the released result files: raw model outputs are stored per item, scoring is exact match, and all resampling is seeded. Local model runs use greedy decoding under a fixed grammar and reproduce to within batching-order float noise at pinned versions (vLLM 0.11). API model results are a dated snapshot: the served checkpoints sit behind provider aliases and carry no determinism guarantee, so their raw outputs are released for re-scoring and the claims that must outlive any snapshot are carried by the local open-weight models.

4 Results

4.1 De-saturation

Table 3 is the paper’s first claim. The obvious objection is that any conjunction gets harder, which is why group-scored MCQ is in the table: matching beats it on ceiling (0.52 versus 0.61), floor (5.4×10^{-7} versus 9.8×10^{-4}), and spread (0.44 versus 0.40), and group-scored MCQ has no Section 4.2.

The difficulty is signal, not noise. Three measurements. First, rankings are preserved (Spearman $\rho = 1.0$ against MCQ) while the paired per-model penalty scales inversely with strength: -0.0385 for the strongest model, -0.0929 , -0.1474 for the weakest, all CIs excluding zero. Amplified difficulty on the same construct looks like this, noise would scramble the order. Second, the choices-only baseline measures 0.049, 0.054, 0.040 against the 0.05 floor of Proposition 1, certifying a style-homogeneous pool on QuALITY. By contrast, choices-only beats a majority baseline in 11 of 12 cases on standard MCQA [4]. Third, group-level closed-book rates are about 1% and statistically identical to group-scored MCQ, so the headroom cannot be reached without the passage.

The difficulty is not format tax. On easy-matching pools (Section 3.3) every model exceeds its own MCQ accuracy, 0.975 versus 0.910 for the best model, and the well-formedness screen retains 72.3% of groups with every headline number stable or improved on the screened subset. The gap from 0.90 exact assignment on separable pools to 0.52 on real pools is discrimination among same-document candidates.

The dials. Sweeping N from 3 to 6 (pools of 12 to 24, floors 7.6×10^{-4} down to 1.0×10^{-8}) moves exact assignment for the best model through 0.680, 0.607, 0.463, 0.367, for the middle model through 0.507, 0.433, 0.273, 0.180, and for the weakest through 0.177, 0.147, 0.060, 0.020, while per-question accuracy barely responds, 0.883 to 0.821 and 0.547 to 0.492. *Difficulty is manufactured by the conjunction, not by making individual questions harder*, which is why the group-scored comparison is the honest one. Model separation is preserved throughout: the paired gap between the two Gemini models is -0.092 , -0.083 , -0.077 , -0.079 across the sweep, statistically indistinguishable, so N changes how hard the benchmark is without changing what it measures. The 26-letter answer alphabet caps distractor-pooled groups at $N=6$. The withhold dial leaves the false-answer rate essentially flat from $p=0.2$ to 0.8 (best model 0.34 to 0.26, weakest 1.00 to 0.99, and the accuracy-tied pair’s gap sits between $+0.36$ and $+0.44$ at every level), so the disposition is a stable per-question property rather than a base-rate response, and p buys observations per group without changing what is measured. High p does stress format compliance for unconstrained API models (up to 151 unparsable responses at $p=0.8$) while grammar-constrained decoding stays at zero.

4.2 Confabulation is rampant and invisible to accuracy

Three observations from Table 4.

Confabulation is near total for some models. Qwen3-8B declined 2 of 900 unanswerable items across both withhold levels, and 0 of 2,100 answerable ones. Under competitive selection with no offered escape, its disposition to answer is absolute.

Accuracy is statistically blind to it. Gemini 3.6 Flash and 3.1 Flash-Lite differ in accuracy by $+0.0022$, 95% CI $[-0.0269, +0.0325]$, and the sign flips between $p=0.2$ and $p=0.4$, so the ordering is noise. Their false-answer rates differ by $+0.4192$, CI $[+0.3754, +0.4646]$. The contrast replicates at $p=0.2$: accuracy -0.0208 $[-0.0475, +0.0050]$, false-answer rate $+0.3967$ $[+0.3367, +0.4600]$. Ability is held constant by measurement, so the confabulation gap cannot be attributed to competence.

Table 3 De-saturation on held-out QuALITY ($N=5$). Same labels, no authoring. Exact assignment restores 39 points of headroom over MCQ for the best model, drops the floor six orders of magnitude, and doubles model separation. Group-scored MCQ (all five items of a group correct) is the honest conjunctive baseline and sits in this table on purpose.

	MCQ	Group-scored MCQ ($\times 5$)	Matching (exact)
Gemini 3.6 Flash	0.9104	0.6067	0.5200
Gemini 3.1 Flash-Lite	0.8761	0.4967	0.3167
Qwen3-8B	0.7153	0.2067	0.0800
Best-to-worst spread	0.20	0.40	0.44
Guessing floor	0.25	9.8×10^{-4}	5.4×10^{-7}
Closed-book rate (group level)	0.0067	0.0067–0.0133	0.0033–0.0133

Table 4 Withhold arm at $p=0.4$ (2 of 5 golds deleted, 600 to 828 unanswerable encounters per model), eight models. Strict exact assignment requires every assignment correct and every unanswerable question declined. Accuracy cannot express the strict column.

	Accuracy	False answer ↓	False abstention	Strict ↑
Gemini 3.6 Flash	0.7500	0.2783	0.1444	0.2367
Gemini 3.1 Flash-Lite	0.7531	0.6970	0.0449	0.0633
Gemma 4 26B	0.6900	0.6767	0.0844	0.0667
Gemma 3 27B	0.6159	0.9179	0.0225	0.0024
Gemma 3 12B	0.5356	0.9700	0.0156	0.0000
Qwen3-32B (FP8)	0.6707	0.9891	0.0008	0.0000
Qwen2.5-7B	0.4267	0.9450	0.0322	0.0000
Qwen3-8B	0.5189	0.9983	0.0000	0.0000

Abstention is a two-error frontier. The model with the lowest false-answer rate buys it with the highest false-abstention rate (0.1444). A single accuracy number collapses this trade-off to nothing.

4.3 The affordance effect

Measured abstention is a joint property of the model and the response affordance (Table 5 and Figure 3). The shift is uniform in direction across all four models and both families. Single-affordance abstention numbers are therefore substantially instrument artifacts, and existing abstention results are reported without controlling for this.

The decomposition is model dependent. Matching and None-MCQ differ in two ways at once, and the middle arm splits them: Matching to Prose-MCQ removes competition at fixed permission, Prose-MCQ to None-MCQ swaps the prose permission for a printed option at fixed independence. For the three instruction-tuned models, competition supplies most of the total false-answer reduction (66%, 87%, 54%). For Qwen3-8B the prose permission is inert, 0.998 to 0.975, two uses in 600 opportunities, and the printed option supplies 96% of its effect. Capable models are silenced by competition, the weakest abstainer does not use permission and needs the escape printed as an option. One bundling caveat: the competition step also changes the number of live candidates (about 17 versus 3), so its share combines exclusivity with candidate count.

On discrimination, removing the printed escape roughly doubles the false-answer spread

Table 5 Three response affordances on identical items with identical withheld questions ($p=0.4$). Escape salience rises left to right: Matching offers a one-line prose permission against roughly 17 live candidates, Prose-MCQ the same permission on independent three-option items, None-MCQ a printed escape option. False answers fall and false abstentions rise with salience, for every model.

	False-answer rate			False-abstention rate		
	Matching	Prose	None	Matching	Prose	None
Gemini 3.6 Flash	0.2783	0.1733	0.1200	0.1444	0.1967	0.2700
Gemini 3.1 Flash-Lite	0.6970	0.2988	0.2379	0.0449	0.1384	0.1644
Gemma 4 26B	0.6767	0.4941	0.3378	0.0844	0.0862	0.1394
Qwen3-8B	0.9983	0.9750	0.4567	0.0000	0.0022	0.1889

Table 6 Eight models across four lines, identical items, $p=0.4$. The competition share is the fraction of the matching-to-None false-answer gap supplied by removing competition, $(\text{Match} - \text{Prose})/(\text{Match} - \text{None})$. Gemma 4 is a sparse model with about 4B active parameters, so its size is not comparable to the dense entries.

	MCQ acc.	False-answer rate			Comp. share
		Match	Prose	None	
Qwen2.5-7B	0.701	0.945	0.930	0.380	0.03
Qwen3-8B	0.715	0.998	0.975	0.457	0.04
Qwen3-32B (FP8)	0.803	0.989	0.767	0.529	0.48
Gemma 3 12B	0.731	0.970	0.807	0.563	0.40
Gemma 3 27B	0.790	0.918	0.604	0.581	0.93
Gemma 4 26B	–	0.677	0.494	0.338	0.54
Gemini 3.1 F-Lite	0.876	0.697	0.299	0.238	0.79
Gemini 3.6 Flash	0.910	0.278	0.173	0.120	0.66

across models (None 0.34, Matching 0.72, Prose 0.80, the last driven by Qwen’s ceiling), while the headline accuracy-tied pair separates most under competition (+0.4192 versus +0.1249 Prose and +0.1179 None).

Two consequences with teeth. First, the accuracy-blindness finding replicates on the second instrument: Flash-Lite and Gemma are accuracy-tied on None-option MCQ (+0.0041, CI [−0.0259, +0.0334]) yet differ significantly in false-answer rate (−0.0999, CI [−0.1376, −0.0619]). Second, **the affordance flips an accuracy ranking outright**. On plain MCQ, 3.6 Flash beats Flash-Lite (−0.0343, CI [−0.0505, −0.0189]). Add the none option and Flash-Lite beats 3.6 Flash (+0.0656, CI [+0.0344, +0.0967]), both significant with opposite signs, because 3.6 Flash’s cautious disposition (false-abstention rate 0.27) consumes its answerable accuracy. Leaderboard positions are already affordance-dependent today, and nobody is controlling for it.

4.4 Scale, generation and family

Two dense scale pairs from independent developers, plus two generation steps, let us separate what scale buys from what training choices buy (Table 6).

Competitive confabulation does not fall with scale. Within each family the matching false-answer rate is essentially flat: Qwen 0.998 to 0.989 across a 4× step, Gemma 3 0.970 to 0.918 across a 2.3× step. This reproduces AbstentionBench’s conclusion that scaling does not fix abstention, on a sharper instrument. Quantization is not the driver, since the 8B ran at bf16 and

Table 7 MMLU-Pro, $N=3$ groups by source topic, pool $M=24$ keeping 7 of each question’s 9 distractors, MCQ mirror-paired to the same questions. No passage exists, everything is closed book. Choices-only floor is $1/24=0.042$ with cap $1/3$.

	Choices-only	Per-question		Group level	
		Matching	MCQ	Match. exact	Grp. MCQ
Gemini 3.6 Flash	0.0975	0.6592	0.7242	0.3125	0.3700
Gemini 3.1 Flash-Lite	0.1012	0.5369	0.6129	0.1950	0.2150
Gemma 4 26B	0.1050	0.4699	0.5955	0.1300	0.2000
Qwen3-32B (FP8)	0.0558	0.3675	0.4492	0.0725	0.1000
Qwen3-8B	0.0667	0.2758	0.3333	0.0275	0.0600

the 32B at FP8 with the same ceiling.

Scale buys prose-following, and this replicates across families. The competition share rises with size in both dense pairs, 0.04 to 0.48 for Qwen and 0.40 to 0.93 for Gemma 3, because the written permission becomes effective (prose false-answer falls 0.975 to 0.767 and 0.807 to 0.604) while the competitive rate does not move. Larger models can act on an instruction to decline. None of them decline when the instruction competes with a full pool of candidates.

Generation is a choice, not a trend. The two developers moved in opposite directions on the same axis. Qwen 2.5-7B to Qwen3-8B made abstention *worse* under every affordance (matching 0.945 to 0.998, prose 0.930 to 0.975, none 0.380 to 0.457), while Gemma 3 27B to Gemma 4 26B made it markedly *better* (0.918 to 0.677, 0.604 to 0.494, 0.581 to 0.338). Abstention disposition is therefore a tunable property of the training recipe rather than an emergent consequence of capability, which is precisely why it needs to be measured and reported rather than assumed.

Family dominates level. The absolute competitive false-answer rate separates the lines far more than scale does within them: Gemini 0.28 to 0.70, Gemma 0.68 to 0.97, Qwen 0.95 to 1.00. A 12B Gemma confabulates more under competition than a Gemini model of any size in our set.

4.5 A second benchmark: de-saturating the de-saturator

MMLU-Pro is the field’s authored answer to saturation: ten options per item, the additions LLM-written, floor 0.1. It has no passage, so groups are source topics and every arm is closed book, and ten-option items overflow the 26-letter alphabet at $N=5$, so the pool keeps 7 of each question’s 9 distractors at $N=3$ ($M=24$). A first pass keeping only 3 of 9 *inverted* the difficulty comparison, matching became easier than MCQ, which localises the transform’s difficulty gain precisely: it lives in the authored distractors the pool retains, and the retention count is a dial the converter must respect on option-rich benchmarks.

With the distractors kept, de-saturation transfers (Table 7). Matching is harder than MCQ per question for all five models (-0.058 to -0.126 , every CI excluding zero, and these contrasts are item-paired since the MCQ arm is mirror-built), the conjunction is harder than group-scored MCQ for every model, the floor is 8.2×10^{-5} against 10^{-3} , and rankings are again preserved ($\rho=1.0$).

The choices-only arm detects MMLU-Pro’s own artifact. On QuALITY this arm sat at the $1/M$ floor. On MMLU-Pro the Gemini-family models clear it decisively (0.098 to 0.105 against 0.042 at $N=3$, 0.078 to 0.080 against 0.050 at $N=5$) while Qwen models stay near it. Writing the leak as separability $\alpha = (\text{observed} - 1/M)/(1/N - 1/M)$, the fraction of the floor-to-cap range achieved, the Gemini family measures $\alpha \approx 0.19$ to 0.22 at *both* N , invariant under the cap change exactly as Proposition 1 predicts for gold-set identification at fixed fidelity, while Qwen measures

Table 8 The transform across four benchmarks. Difficulty is the paired matching-minus-MCQ per-question range across models (every interval’s CI excludes zero). α is Gemini-family partial-input separability, significant only where marked. False-answer rates are the best and worst abstainer at $p=0.4$.

	Domain	Difficulty	α	FA best / worst
QuALITY	long-form reading, open book	−0.04 to −0.15	≈ 0	0.28 / 1.00
RACE	short-passage exams, open book	−0.06 to −0.25	0.17–0.34	0.22 / 0.98
MMLU-Pro	general knowledge, closed	−0.06 to −0.13	0.19–0.22	0.45 / 1.00
C-Eval	subject exams, Chinese	−0.05 to −0.20	≤ 0.11	0.38 / 1.00
GPQA	expert STEM, closed	−0.07 to −0.14	≈ 0	0.67 / 0.99

$\alpha \leq 0.09$. The likely source is MMLU-Pro’s construction, original golds beside LLM-generated added distractors, the artifact of Balepur et al. [4] reproduced inside the de-saturation benchmark, though memorisation of the gold strings would produce the same signature. Either way the transform detects it, caps it at $1/N$, and quantifies it on a calibrated scale. The uncapped version of the same probe on the same items, single-item choices-only, reaches 0.226 to 0.283 against its 0.10 floor for every model including both Qwens, 2.3 to 2.8 times chance under a ceiling of 1.0, while pooling holds every model to at most 0.105 under the provable ceiling of $1/3$. Notably, Qwen exploits the single-item cue as strongly as the Gemini family yet measures $\alpha \leq 0.09$ in the pool, so pooling does not merely cap the artifact: mixing distractor sets across questions disrupts cues that survive inside one item’s authored option set. The well-formedness screen retains 83.5% of MMLU-Pro groups, flagging only 2 of 1200 questions for a second valid gold, and α is unchanged on the screened subset (0.175 to 0.215), so the leak is separability rather than ambiguity.

Closed-book confabulation is far worse for everyone. On the withhold arm ($p=0.4$, first-pass pools), false-answer rates are 0.448 (3.6 Flash), 0.851 (Flash-Lite), 0.847 (Gemma), 0.981 (Qwen3-32B), 0.997 (Qwen3-8B), all far above the same models’ open-book QuALITY levels, with the Qwen ceiling replicating on a second dataset. Without a passage to check against, even the most disciplined model confabulates on nearly half of unanswerable items, which is the closed-book hallucination case deployment actually faces.

4.6 Generality: four benchmarks, two languages

Three regularities hold across all five benchmarks (Table 8 and Figure 5). **Difficulty is universal:** matching is harder than MCQ for every model on every benchmark, item-paired by mirror construction on RACE, GPQA, C-Eval and MMLU-Pro, and the conjunction is harder than group-scored MCQ wherever the pool keeps the authored distractors. **Separability tracks authoring incentive, not authorship:** α sits at the floor on QuALITY and GPQA, whose distractors were written adversarially by validated crowdworkers and by domain experts, is marginal on C-Eval, and is substantial both on MMLU-Pro, whose added distractors are LLM-generated, and on RACE, whose school-exam distractors are written to be identifiable by students. Human authorship alone does not buy a clean pool. What the instrument measures is how far golds stand apart from distractors, on a scale bounded below by $1/M$ and above by $1/N$, and that quantity varies by a factor of four across benchmarks that all look equally respectable from the outside. **Confabulation escalates with domain difficulty:** the best abstainer’s false-answer rate climbs from 0.22 and 0.28 (open-book reading) through 0.38 and 0.45 (closed-book exams and knowledge) to 0.67 (expert STEM), while the Qwen family stays at 0.93 to 1.00 everywhere.

The affordance decomposition replicates on a second open-book corpus. On RACE the competition step supplies 0.90 to 1.01 of the false-answer gap for the four larger models and 0.12

to 0.16 for the two small Qwens, the same split found on QuALITY (Section 4.3), and it appears on passages of a few hundred words rather than a few thousand. The suppression of abstention under competition is therefore a property of the response format, not of the model losing track of a long document.

C-Eval also produces the first ranking divergence of the study ($\rho=0.90$): MCQ scores Qwen3-8B and Gemma as statistically tied (+0.016, CI [-0.013, +0.045]) while matching separates them decisively toward Gemma (-0.054, CI [-0.085, -0.023]). C-Eval is the benchmark most plausibly memorised by a Chinese-trained family, single-item MCQ is where memorised cues pay, and Section 4.5 showed pooling disrupts cues that survive inside one item's option set, so the collapse of Qwen's MCQ edge under pooling is consistent with pooling acting as a contamination probe, though we do not claim contamination is established.

4.7 Anchoring against generation

The same questions asked open-ended, judged ANSWER versus ABSTAIN (0 of 6,000 unjudged): confabulation 0.0433 (Gemma), 0.0550 (3.6 Flash), 0.0583 (Flash-Lite), 0.3250 (Qwen3-8B), over-refusal 0.0922, 0.0489, 0.0556, 0.0133. Qwen's unwillingness to abstain is model-real, worst on every instrument. A second judge from a different family agrees with the first on 94.9 to 98.8% of the 6,000 classifications and reproduces every rate to within 0.01, so the criterion does not depend on the judge.

At the item level (2,389 paired unanswerable items), matching's false-answer signal agrees with open-ended confabulation about twice as well as the printed-option instrument's, pooled ϕ 0.181 versus 0.096, paired difference +0.085, CI [+0.039, +0.129]. All four per-model differences are positive though individually not significant, and the pooled figure includes between-model separation as well as within-model item agreement, so we read this as directional support for the competitive affordance rather than a settled verdict. Both absolute agreements are modest, partly because the criterion is ceiling-limited: a wholly mismatched passage is easy to detect, and the three Google models sit within about 1.5 SE of one another on it.

4.8 Controls

Table 9 Controls, one line each. All run.

Control	Result
Group-scored MCQ	in Table 3, de-saturates too, no abstention axis
Choices-only	provably chance, measured 0.049/0.054/0.040 vs 0.05
Easy-matching	every model exceeds its own MCQ accuracy, format tax is zero
Verifier screen	72.3% retention, headline numbers stable or improved on the screened subset
Closed-book filter	from closed-book runs only, transfers across families
Constrained decoding	no unparsable responses, unguided control moves false-answer rate by 0.004
Judge scope	ANSWER/ABSTAIN only, 0/6,000 unjudged, second judge agrees 94.9 to 98.8%
Item alignment	all instrument arms identical items, verified 2,070/2,070

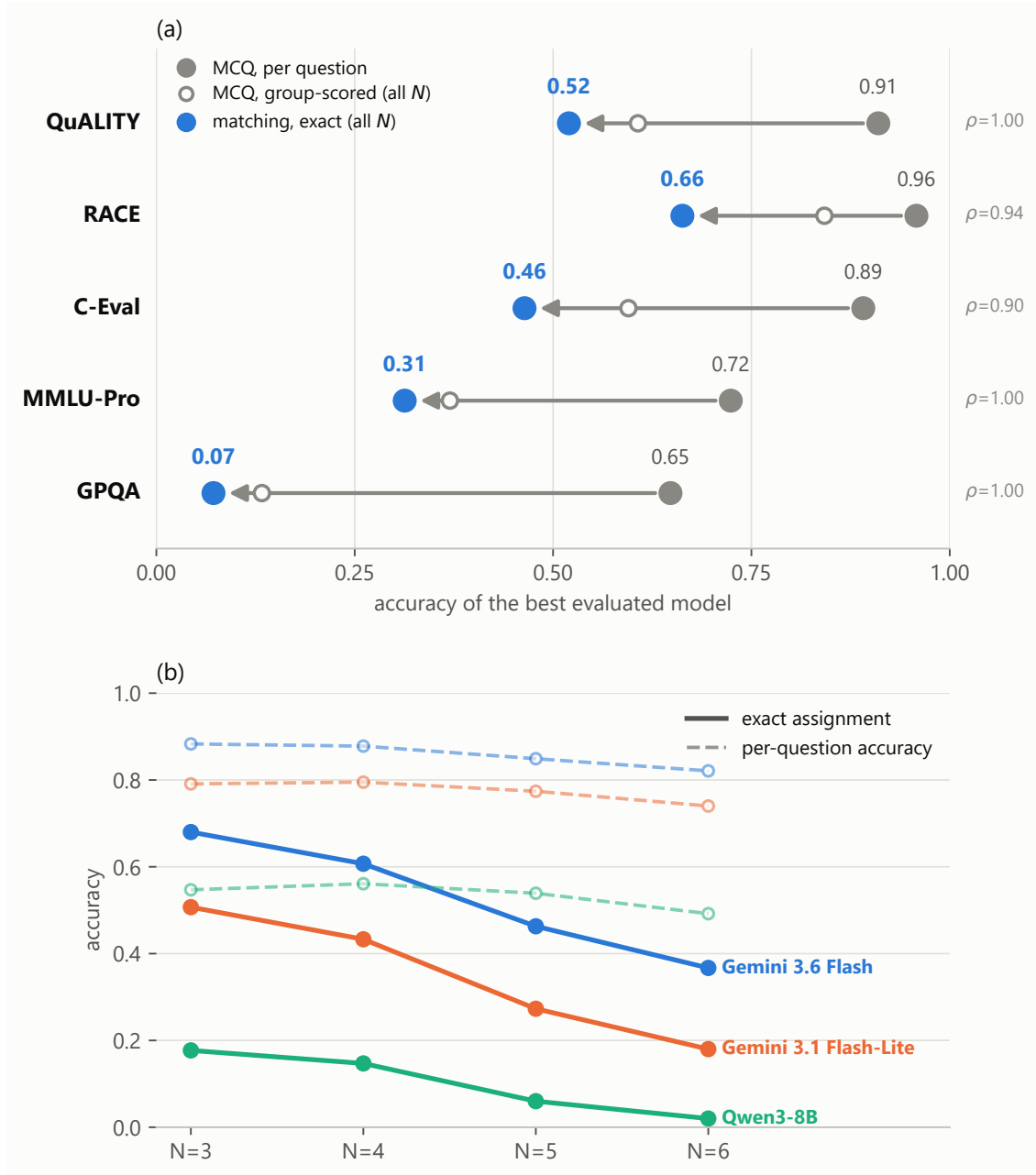


Figure 2 De-saturation at a glance, and the dial that produces it. (a) On five benchmarks the best evaluated model moves from its per-question MCQ accuracy (gray) through group-scored MCQ (hollow, the same all- N conjunction applied to MCQ) to matching exact assignment (blue), with model rankings preserved or nearly preserved (ρ per row). Guessing floors fall from 0.25 (0.10 on MMLU-Pro) to between 8×10^{-5} and 5×10^{-7} . (b) The N dial on QuALITY: exact assignment (solid) falls steeply from $N=3$ to 6 while per-question accuracy (dashed) barely moves, so difficulty is manufactured by the conjunction rather than by harder individual questions, and the paired gap between the two Gemini models is statistically unchanged across the sweep.

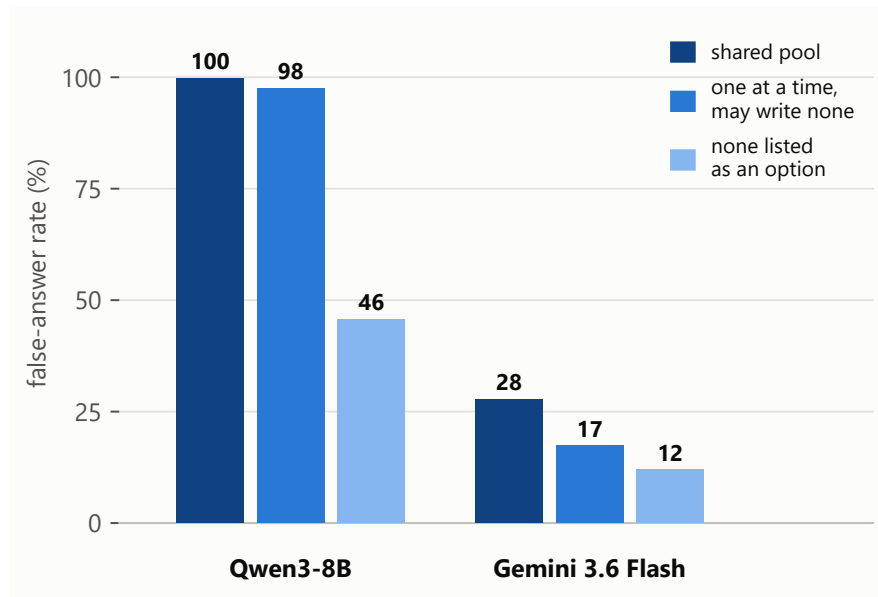


Figure 3 The same unanswerable questions asked three ways. False-answer rates for the weakest and strongest abstainer under the three response affordances of Table 5: candidates in one shared pool, questions one at a time with a prose permission to write none, and none printed as a lettered option. Escape salience rises left to right within each group and every model’s rate falls with it.

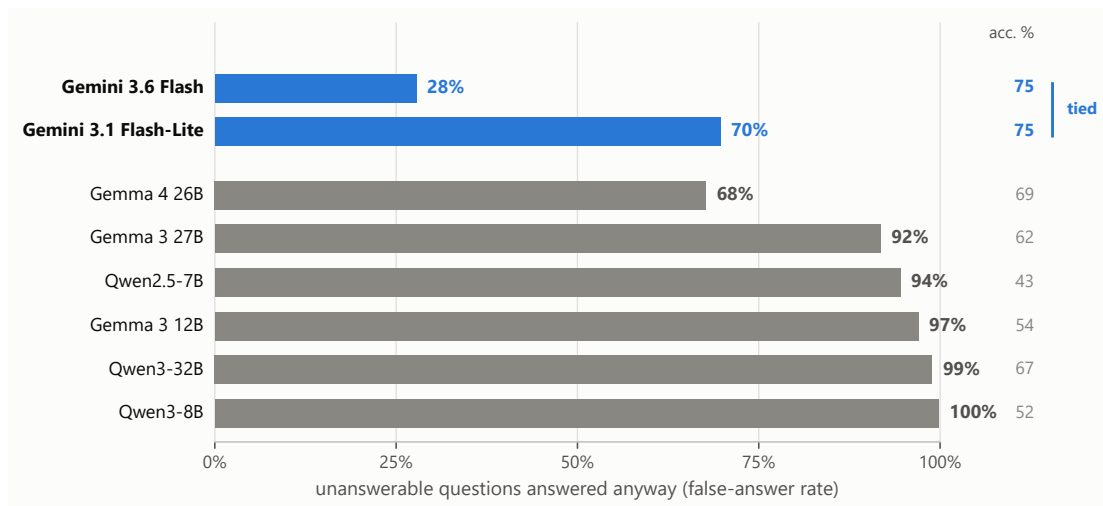


Figure 4 Eight models on the QuALITY withhold arm ($p=0.4$): the fraction of unanswerable questions each model answered anyway, with open-book accuracy on the same arm in the right column. The two highlighted models are statistically tied on accuracy ($+0.002$, CI includes 0) yet 2.5× apart on false answers ($+0.42$, CI $[+0.38, +0.46]$). Qwen3-8B declined 2 of 900.

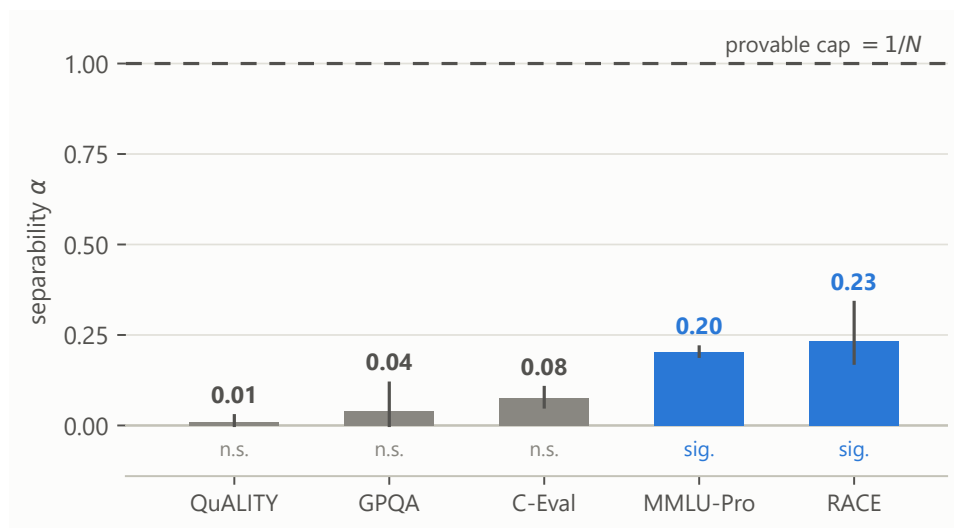


Figure 5 The separability detector across five benchmarks. Bars are the Gemini-family mean of $\alpha = (\text{observed} - 1/M)/(1/N - 1/M)$ on the choices-only arm, whiskers the per-model range, the dashed line the provable $1/N$ cap of Proposition 1. Blue bars are significantly above the $1/M$ floor. The instrument clears the adversarially authored pools (QuALITY, GPQA), and flags the LLM-written (MMLU-Pro) and school-exam (RACE) distractor sets.

5 Discussion

What the transform buys, precisely. Two things, and the wording matters. First, restored headroom and discrimination on existing labels: a 39-point ceiling lift, a floor below 10^{-6} , doubled spread, preserved rankings. Second, an abstention axis with exact free ground truth from the same forward pass. We deliberately do not claim the format measures more comprehension per point. Our own passage-contribution analysis rules that phrasing out (Section ??).

Why the affordance finding matters beyond this paper. Abstention studies report one number per model. Our paired arms show that number is a joint property of model and response affordance: the same model moves from a 0.998 to a 0.457 false-answer rate when a none option is printed, and every model moves in the same direction. Any two abstention results measured under different affordances are therefore not comparable, and a leaderboard that adds an escape option is not a harder version of the same leaderboard but a different ordering. We observed a significant accuracy rank flip from the affordance alone.

A ladder that forks at the top. The instruments order naturally by selection pressure. Plain MCQ applies pressure but cannot observe abstention. Delete-gold MCQ observes it under a printed escape. The pool observes it under maximal pull, roughly 17 live candidates and no printed escape. Measured confabulation tracks the pressure for every model: generation 0.04 to 0.33 (under an easier unanswerability condition), printed escape 0.12 to 0.46, competition 0.28 to 1.00. The rungs are therefore not interchangeable measurements of one quantity. We climb to the top rung for three measured reasons and one argued reason. Only the pool de-saturates the benchmark. The pool separates models about twice as widely on the false-answer rate (spread 0.72 versus 0.34, headline pair +0.419 versus +0.118). Strict exact assignment is hardest and least saturated there. The decomposition arm splits the ladder's top step by mechanism, and the split is itself a model property that moves with scale: the competition share runs from 0.03 at 7B to 0.93 at 27B across eight models and four lines (Section 4.4), so the rungs differ in mechanism as well as magnitude and small models are governed by the printed option where large ones are governed by competition. Deployment prints no escape option, so the competitive affordance is the ecologically relevant one, an argument rather than a measurement. And the external anchor now leans the same way: at the item level the competitive rung agrees with open-ended confabulation about twice as well as the printed-option rung (pooled ϕ 0.181 versus 0.096, paired difference significant, Section 4.7), directional support rather than a settled verdict. The disagreement between rungs remains itself the finding.

Adoption cost. A leaderboard gains an abstention column for the cost of a dataset transform. No judge, no new labels, no second evaluation run. The conversion applies to any benchmark with several questions per context, and the three dials (N, M, p) let the same annotations serve difficulty levels from screening to frontier.

Construct scope. The withhold arm measures detection that the answer is absent from an offered set, reachable partly by elimination. It is not epistemic humility. A model can know the answer and still be scored on noticing its absence. This is the SQuAD 2.0 construct manufactured for free, not a claim about knowing what one does not know.

6 Conclusion

Saturated MCQ benchmarks do not need new items, they need a harder question asked of the items they already have. Pooling the answers of co-document questions into one exclusive candidate list lifts the ceiling by 39 points, drops the guessing floor six orders of magnitude, and doubles model separation, with rankings preserved and a provably chance partial-input

baseline. The same pool then funds a measurement shipped benchmarks do not make: deleting golds manufactures unanswerable questions from existing labels, and running the deletion under two paired affordances on identical items shows measured confabulation is rampant, invisible to accuracy, and strongly instrument dependent, to the point of flipping a significant accuracy ranking. Both payoffs come from one transform that authors nothing, needs no judge, and returns abstention from the same forward pass as accuracy. We release the conversion pipeline so that any grouped MCQ benchmark can be revived, and audited, for free.

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